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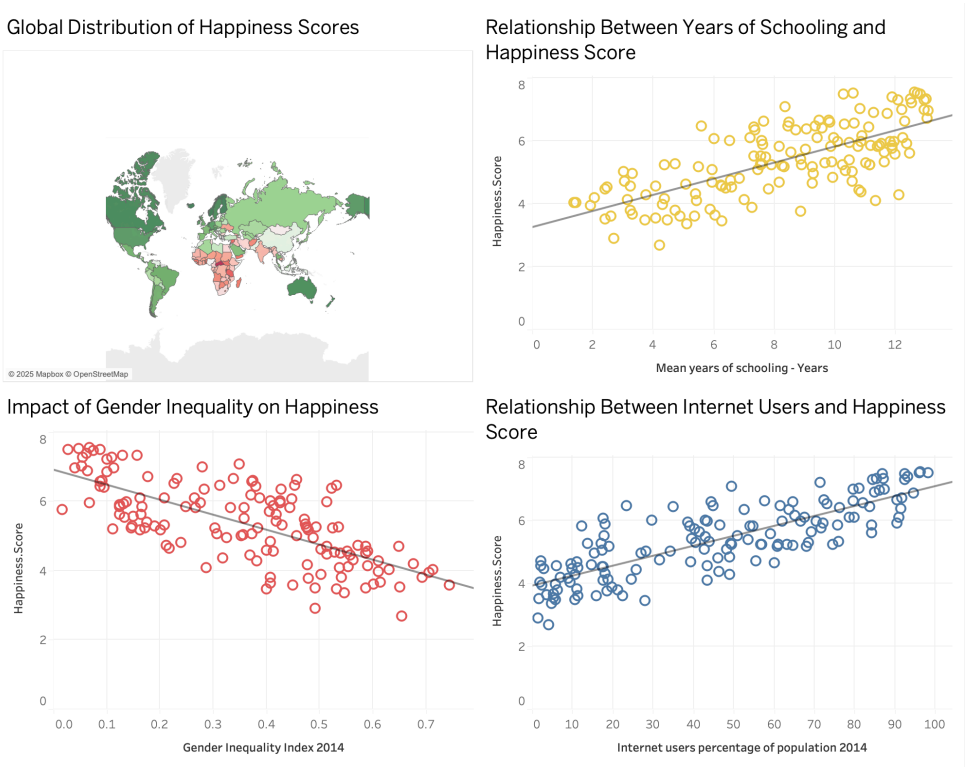
Tableau Assignment 1

Challenges Faced While Creating the Dashboard

1. Driver Compatibility Issues: One of the main challenges I faced was downloading and configuring the MariaDB drivers. Since the driver was not compatible with macOS, I was unable to establish a direct database connection. As a workaround, I decided to use the CSV files for analysis and visualization instead.
2. Understanding Data Joins: While importing multiple datasets, I was initially unsure about which type of join to use and where to apply it within the tool. After some research, I learned that an inner join works by default, and that I could join the tables using common fields such as ID and Country to merge the data effectively.
3. Map Color Grading: Another challenge was implementing color gradients on the map to visually represent the happiness scores. I struggled to adjust the light-to-dark effect that indicates variations in the data. After discussing the issue with my professor, I gained a better understanding of how to configure the color scale properly to enhance the visualization.



1. Global Distribution of Happiness Score:

The map visualization provides a global overview of happiness levels across countries. The green-to-red color gradient represents how happiness varies geographically—countries shaded in darker green show higher happiness scores, while those in red indicate lower scores.

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2. Years of Schooling vs. Happiness Score

Correlation: Positive
Insight: As the *mean years of schooling* increase, the *happiness score* also rises. Countries with better access to education and longer average schooling years tend to report higher happiness levels.
Interpretation: Education contributes to happiness by enabling better employment, financial stability, and informed lifestyle choices. It also fosters social awareness and empowerment.
Summary: Education acts as a foundational driver of well-being—improving quality of life, opportunity, and optimism.

3. Gender Inequality vs. Happiness Score

Correlation: Negative
Insight: Countries with *lower gender inequality* (lower index values) show *higher happiness scores*. The trend suggests that societies promoting gender equality experience greater collective well-being.
Interpretation: Gender equality reflects inclusivity and fair access to opportunities, which strengthen social trust and life satisfaction. High inequality often signals restricted opportunities and systemic barriers, lowering happiness levels.
Summary: Reducing gender inequality enhances happiness, as balanced societies provide fairness, respect, and shared prosperity for all genders.

4. Internet Users vs. Happiness Score

Correlation: Positive
Insight: Higher percentages of *internet users* correlate with higher *happiness scores*. Countries with better digital connectivity tend to have happier populations.
Interpretation: Internet access improves communication, education, access to services, and global awareness—factors that directly and indirectly boost quality of life. However, digital access alone does not guarantee happiness; the quality and purpose of use also matter.
Summary: Digital inclusion supports social and economic participation, contributing to higher well-being in connected societies.

Conclusion:

This dashboard presents a comparative analysis of the factors influencing national happiness levels across countries. It combines a global map of happiness scores with scatterplots showing relationships between happiness and key socio-economic indicators **education (years of schooling)**, **gender inequality**, and **internet usage**.
The visualizations reveal that **higher education levels and internet access** are positively correlated with happiness, while **greater gender inequality** is associated with lower happiness scores. Together, these insights highlight how social development, equality, and digital connectivity contribute significantly to the overall well-being of a nation.

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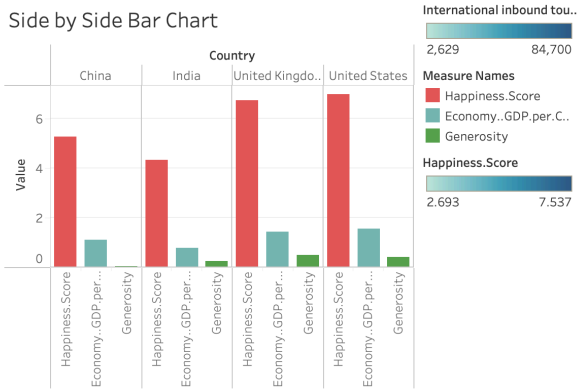
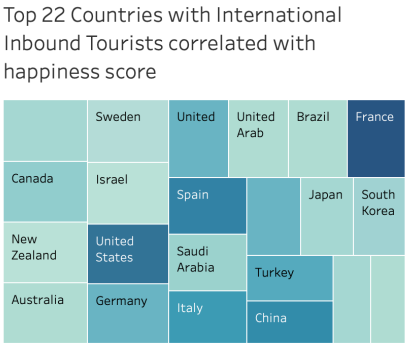
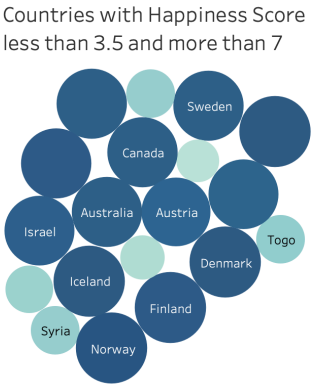
Extra Dashboard and Visualizations:







1. **List of Countries by Happiness Score**
   Type: Table
   Description: This table displays countries ranked by their overall happiness score. It allows quick comparison of the top-performing nations, with **Norway, Denmark, and Iceland** among the happiest countries globally.
2. **Side-by-Side Bar Chart**
   Type: Bar Chart
   Description: This chart compares multiple indicators: **Happiness Score, GDP per capita, and Generosity** across selected countries (China, India, UK, USA). It helps visualize how economic prosperity and generosity vary alongside happiness.
3. **Top 22 Countries with International Inbound Tourists Correlated with Happiness Score**
   Type: Tree Map
   Description: The treemap shows the top 22 countries attracting the highest numbers of international tourists. The color intensity represents the **happiness score**, suggesting that **countries with higher happiness levels often attract more tourists**.
4. **Countries with Happiness Score Less Than 3.5 and More Than 7**
   Type: Bubble Chart
   Description: This bubble chart categorizes countries into two groups — those with **low happiness (below 3.5)** and **very high happiness (above 7)**. The bubble size highlights each country's score, making it easy to spot extremes like **Norway, Finland, and Syria**.